EXCAVATE

PROBLEM STATEMENT

2026

COMPOSIT | 31ST EDITION

MATRISK AI: PREDICTING COMMODITY MARKETS USING MATERIAL SCIENCE SIGNALS

OVERVIEW

Modern financial markets and industrial supply chains depend heavily on the physical properties of materials such as steel, copper, lithium, and rare earth elements. However, most financial and commodity prediction models operate independently of the underlying material science governing these assets.

This challenge introduces **MatRisk AI**, a cross-domain modelling problem that combines **materials science**, **commodity markets**, and **machine learning** to generate predictive insights. Participants must develop data-driven models that **integrate material properties with commodity market dynamics** to uncover predictive signals and improve market forecasting.

The objective is to design models that leverage **material science intelligence and financial time-series analytics** to better understand and predict commodity behaviour.

DATASET DESCRIPTION

Participants will work with the following datasets - *Dataset Link* + *Reference Guide*

DS1: Material Properties Dataset
Contains 5,500 materials with structural and physical properties.
Key columns include:

- Material formula and crystal structure
- Band gap
- Shear modulus
- Density
- Formation energy
- Bulk modulus
- Poisson's ratio
- Melting point

This dataset represents composition, structure and property relationships used in computational materials science.

Physical constraints which ensure that predictions remain **physically plausible -**

- Elastic relationship:

$$E = 2G\,(1+\nu) = 3K\,(1-\nu)$$

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- Material stability constraints
  - formation_energy < 0 for stable compounds
  - $-1 < \nu < 0.5$
  - bulk_modulus > 0
  - shear_modulus > 0
  - density > 0

DS2: Commodity Price Dataset

Contains 10 years of daily price data for 8 industrial commodities, including -

- Steel HRC
- Copper
- Lithium
- Rare Earth Index
- Aluminium
- Nickel
- Cobalt
- Iron Ore

The dataset includes engineered financial indicators such as -

- Daily returns
- Moving averages
- MACD
- Momentum indicators
- Rolling volatility
- RSI
- Bollinger bands
- Term spread

The dataset spans approximately 2,870 trading days per commodity (2014–2024).

DS3: Cross-Domain Material–Financial Features

This dataset bridges materials science and financial markets. Participants must merge DS3 with DS2 to generate enriched financial signals.

Key features include:

- Material Quality Index (MQI)
- Supply disruption probability
- Green premium
- Market concentration (Herfindahl) index
- MQI trend indicators
- Substitution elasticity
- Carbon intensity

DS4: MQI Weights

This dataset gives the weightage of different factors in the calculation of Material Quality Index of a material.

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TASKS

Task 1: Material Property Modelling

Using DS1, develop models that learn relationships between material composition, structural characteristics, and physical properties, and use these insights to estimate the Material Quality Index (MQI) for the given materials.

Participants are encouraged to explore data-driven, physics-informed, or hybrid modelling approaches to capture the underlying material behaviour.

Expected outcomes -

- Mechanical, Thermal and Electronic properties
- Material Quality Index

Task 2: Commodity Price Prediction

Using DS2 and DS3, develop models to explore and forecast commodity price movements while examining how signals from material science and supply dynamics influence financial markets. Participants may employ time-series analysis, machine learning techniques, or hybrid approaches to capture relationships between material indicators and market behaviour.

Questions to address -

- Do changes in material quality predict commodity prices?
- Can supply disruption probabilities anticipate market volatility?
- How does substitution elasticity affect commodity demand cycles?

Bonus Task: Cost-Aware Inverse Material Design

In this optional challenge, teams will explore inverse material design by proposing new material compositions that achieve strong predicted performance while remaining economically feasible.

Candidate alloys or compounds (e.g., Fe, Ni, Cu, Li, Co, Nd) should be evaluated using the property prediction models from Task 1 to estimate properties and compute the Material Quality Index (MQI).

Using the **element price data** in **DS5**, teams should then estimate the economic cost of each candidate material. The objective is to identify materials that provide a favourable **performance-cost trade-off**.

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DELIVERABLES

Teams should submit -

- A .ipynb notebook
- A trained predictive model
- Report explaining
  - Methodology
  - Feature Engineering
  - Model Architecture
  - Evaluation Metrics
- A Video explaining the solution
- Visualisations or insights derived from the models

SUBMISSION RULES

- Report Slide Limit - 10 slides
- Report Submission Format - .pdf only
- Video Time Limit - 5 mins
- Video Submission Format - .mp4 only
- Submission Link - *Excavate (Submission Round)*
- First slide of the presentation must include -
  - Team Name & Team ID
  - Members' Names & COMPOSIT IDs

EVALUATION CRITERIA

Submissions will be evaluated based on:

- Model performance (40%)
- Innovation in feature engineering (20%)
- Integration of material science and financial data (15%)
- Interpretability and insights (15%)
- Code quality and reproducibility (10%)

CONTACT

DEBARSHINI MONDAL - 9933163116
JAYESH ANDHALE - 9767922637